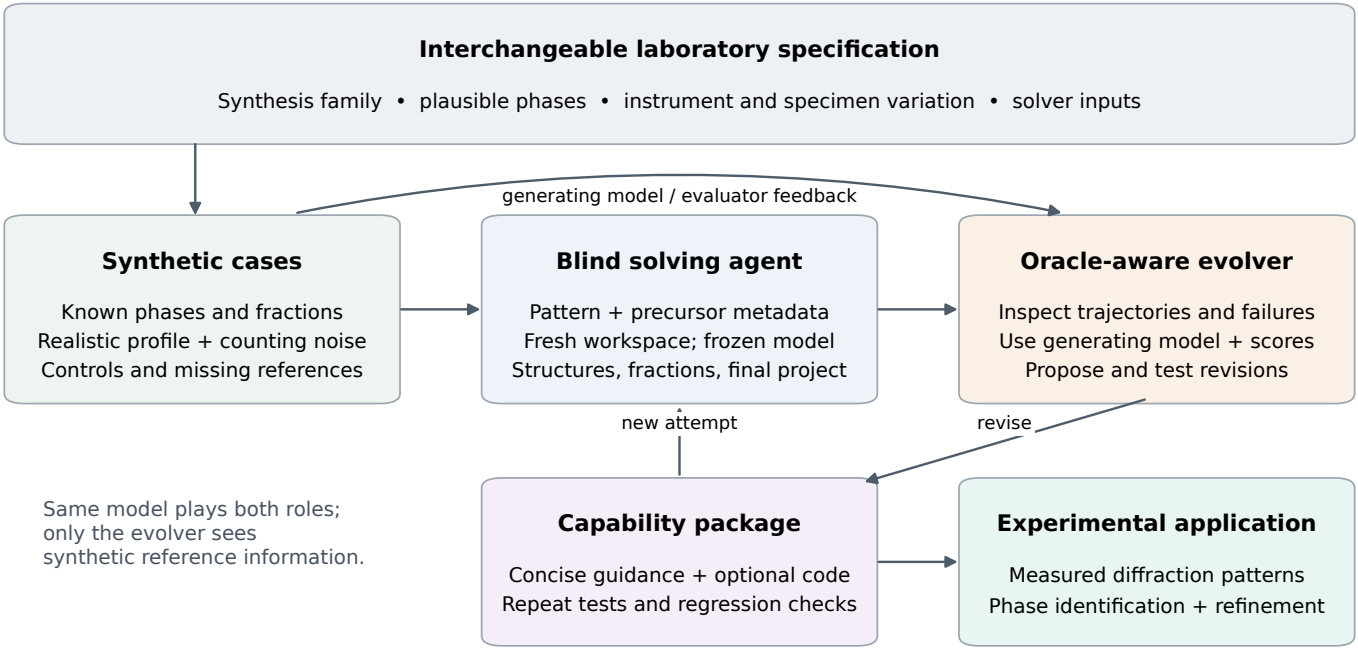
Interchangeable laboratory specification
Synthesis family • plausible phases • instrument and specimen variation • solver inputs
generating model / evaluator feedback
Synthetic cases
Known phases and fractions
Realistic profile + counting noise
Controls and missing references
Blind solving agent
Pattern + precursor metadata
Fresh workspace; frozen model
Structures, fractions, final project
Oracle-aware evolver
Inspect trajectories and failures
Use generating model + scores
Propose and test revisions
new attempt
revise
Same model plays both roles;
only the evolver sees
synthetic reference information.
Capability package
Concise guidance + optional code
Repeat tests and regression checks
Experimental application
Measured diffraction patterns
Phase identification + refinement